



# Muhammad Minto Aji Shodiq

Electrical Engineer · Industrial Maintenance & Panel Specialist

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## PROFESSIONAL SUMMARY

Electrical Engineer with 6+ years of continuous engagement across 4 industries — WWTP/WTP EPC, railway manufacturing (INKA Group), industrial automation R&D, and engineering services — covering electrical system design, panel installation, PLC/HMI, VSD/VFD, SCADA, instrumentation, commissioning, and preventive & corrective maintenance. Proven track record executing complete electrical works independently across 8 industrial facilities nationwide (Java, Sumatra, Batam), covering panel commissioning, vendor coordination, QA/QC documentation, and compliance with K3/PUIL electrical safety standards. CAD-certified (BNSP), experienced in drafting Machine Work Instructions and technical reports, and consistently available for 24/7 on-call field deployment.

## EXPERIENCE

### Electrical & Instrumentation Engineer

Dec 2024 – Apr 2026

**PT Panca Jasa Lingkungan (PJL Enviro)** · WWTP/WTP EPC — Engineering, Procurement & Construction

- Served as sole on-site engineer — without direct supervision — overseeing all M&E maintenance activities across 8 industrial WWTP/WTP facilities nationwide (Java, Sumatra, Batam), covering pumps, blowers, motors, VSD/VFD, electrical panels, PLCs, HMI, and instrumentation.
- Installed, commissioned, and maintained MCC panels, MV/LV distribution panels, cable trays, and electrical wiring systems for industrial wastewater treatment plants; performed energization and system verification as lead commissioning tester.
- Designed electrical systems — cable sizing, load calculations, transformer specs, MCC, and MV/LV distribution — for new-build STP facilities (RS Serpong, Lamongan), producing detailed engineering documentation aligned with project specifications.
- Prepared daily, weekly, and monthly technical status reports and QA/QC documentation (CX schedules, testing reports, Method of Statements/MOS) submitted to project management.
- Coordinated with multiple external vendors on electrical and non-electrical works to ensure timely execution of maintenance and shutdown plans, resolving field technical issues independently.
- Enforced K3 safety protocols and PUIL electrical codes across all site activities — achieving zero Lost Time Injury (LTI) throughout 16 months of multi-site field deployment.
- Diagnosed and resolved critical field failures — including SCADA faults, VSD trips, PLC errors, sensor malfunctions (DO sensors, flowmeters, ultrasonic level sensors 4-20mA), and motor breakdowns — responding on-call across multiple geographically dispersed site locations.

### Hardware & Electrical Trainer

Jan 2024 – Nov 2024

**PT Stechoq Robotika Indonesia** · R&D & Industrial Automation · Yogyakarta

- Developed and delivered BNSP-curriculum-based technical training programs covering PLC programming, VFD systems, sensor integration, electrical wiring, and industrial control systems for engineering trainees — creating structured Machine Work Instructions and training materials.
- Supervised component installation, wiring, and PLC/microcontroller programming during production, ensuring adherence to quality and safety standards.
- Coordinated PLC-HMI and servo/VFD integration projects, overseeing production quality during system build and verification.

### Hardware Engineer — R&D Division

Aug 2023 – Dec 2023

**PT Stechoq Robotika Indonesia** · R&D Company · Industrial Automation · Yogyakarta

- Acted as Project Manager for an automated sheet-metal bending machine integrating PLC-HMI, an ARM Robot assembly machine, and VFD-controlled 3-phase motors — directing project scope, team execution, and full system commissioning from concept to delivery.
- Programmed PLC automation logic and VFD parameters for industrial machine control; performed complete electrical panel wiring and component installation.

### Electrical Engineer — Technology & Operation Dept.

Aug 2022 – Jul 2023

**PT Rekaindo Global Jasa (INKA Group)** · Railway Engineering Consultant · Madiun

- Coordinated the design, fabrication, and quality inspection of KRDE control panels, static frequency inverter panels, and main electric motor panels for KRL commuter trains — ensuring panels met engineering specifications before production release.

- ▶ Planned and finalized electrical design schematics for train propulsion and auxiliary systems; conducted routine maintenance and quality control inspections on production machinery, ensuring all outputs met engineering specifications prior to release.
- ▶ Led innovation research on reengineering the 1,500 Vdc DC-to-DC power conversion system for DC Chopper-based electric trains.

**Mechanical & Electrical Service**

Jan 2021 – Apr 2021

**PT Area Putra Kencana Sejahtera** · *Building Electrical Services · Yogyakarta*

- ▶ Installed, inspected, and maintained building electrical distribution panels and indoor chiller systems, ensuring operational continuity of building M&E infrastructure.

**Electrical Engineer (Freelance)**

Nov 2019 – Present

**Independent / Personal Practice** · *Electrical Design, Installation & Coaching*

- ▶ Designed and installed electrical systems for residential, commercial, and industrial facilities — including load calculation, single-line diagram preparation, circuit breaker sizing, cable sizing, and BoQ preparation.
- ▶ Designed electrical control and remote automation systems for small industrial applications.
- ▶ Developed Excel-based systems for KPI tracking, financial reporting, and engineering dashboards — delivering structured data management tools used across accounting and operational reporting workflows.
- ▶ Served as lead coach for student teams in LKS SMK (Vocational Student Competency — Electrical Installation) at DIY Provincial level for 6 consecutive years (2021–2026), consistently developing participants in panel installation, PLC programming, and smart electrical systems.

**KEY TECHNICAL CAPABILITIES**

**Technical / Hard Skills**

|                                            |                                       |                                        |
|--------------------------------------------|---------------------------------------|----------------------------------------|
| Electrical Panel Installation (MCC, MV/LV) | PLC Programming & HMI Integration     | VSD / VFD Installation & Setting       |
| AutoCAD 2D / 3D (BNSP Certified)           | SCADA & Instrumentation Systems       | BoQ, Cable Sizing & Load Calculation   |
| Preventive & Corrective Maintenance        | Commissioning & Testing (Lead Tester) | K3 / OSH / PUIL Compliance             |
| Technical Documentation (MOS, CX, Reports) | Machine Work Instruction Writing      | MS Office (Excel Reports & Dashboards) |

**Soft Skills**

|                               |                               |                            |
|-------------------------------|-------------------------------|----------------------------|
| Leadership & Team Development | Cross-functional Coordination | Analytical Problem-Solving |
| Adaptability & Agility        | Mentoring & Coaching          | Multi-vendor Management    |
| Independent Decision-Making   | Communication & Reporting     | Project Management         |

**EDUCATION**

**Bachelor of Engineering — Electrical Engineering (S1)**

Aug 2020 – Aug 2024

**Universitas Negeri Yogyakarta (UNY)** · *GPA: 3.74 / 4.00*

- ▶ 3rd Place — Shell Eco-marathon Asia Region 2022, Future Rider Category (International)

**Vocational — Electrical Power Installation Engineering**

Jul 2017 – Jun 2020

**SMK Negeri 2 Yogyakarta** · *TITL — Teknik Instalasi Tenaga Listrik*

- ▶ 2nd Place — LKS SMK DIY 2019: Electrical Installation with PLC (Provincial Level)

**CERTIFICATIONS**

- ▶ BNSP Drafter 3D CAD — K3/OSH, quality procedures, technical measurements, 2D/3D drawing · Mar 2024 – Mar 2027
- ▶ BNSP Vocational Instructor (Pengajar Vokasi) — Curriculum design, K3, training delivery, mentoring & evaluation · Aug 2024 – Aug 2027
- ▶ LKS SMK DIY — Electrical Installation: manual & automatic power control panels, PLC, HMI, Smart Home, KNX · Nov 2019